Cardiac period estimation from non-gated helical CT: sampling criteria and exploratory evaluation

Abstract

BACKGROUND: Cardiac motion undulates the mediastinal border in a non-gated helical chest CT; a recoverable period would let an existing scan be sorted by phase.

OBJECTIVE: To state the sampling conditions that make the period recoverable, test them against real acquisitions, and measure how far an image-based estimate can be trusted.

METHODS: The cycles needed were measured over 144 simulated conditions, evaluated against 192 real series through their headers alone, and tested on 17 image series under a criterion frozen in advance.

RESULTS: Cycles written and samples per cycle multiply to L/dz, so speed only divides a fixed budget. The threshold is 2.5 written cycles, 1.5 for a matched estimator — empirical, not universal. At an assumed cardiac extent, 67 per cent of real series meet the requirements below 100 bpm. On images, 3 of 17 met the self-consistency criterion, yet their cost minima are no deeper than the rejections' and their levels prefer rates differing by 44 to 59 bpm.

CONCLUSIONS: The requirements are computable from headers before any image is analysed. Self-consistency alone does not establish that a period was found: a fit needs the depth of its minimum and a check on what it fitted.

Keywords

Computed tomography; Cardiac imaging techniques; Image processing, computer-assisted; Signal processing, computer-assisted; Sampling studies; Reproducibility of results

1. Introduction

A coronal or sagittal reformat of a non-gated helical chest CT shows the left mediastinal border as an undulating line. It is usually read as a motion artefact. Part of it is written by the heart: the table advances while the heart moves, so cardiac motion contributes to the shape of that line whether or not an electrocardiogram was connected.

That is what makes the question in this paper worth asking, and the step from it to a use is a hypothesis rather than a result. If a component attributable to cardiac motion can be separated from the rest of the line, then the spacing of its successive maxima would be a cue to the cardiac period, and its maxima and minima, falling on opposite parts of the cycle, could serve as cues to phase; a scan already acquired might then be sorted by phase and reformatted accordingly, and further along that road used for ventricular volumetry. Such a route would need no electrocardiographic hardware, no additional acquisition and no repeated overlapping rotations over the heart. This paper examines one point on that route — whether the period can be recovered — and examines nothing beyond it. Estimating phase without first estimating a period is a different route and is not studied here either. Phase identification, phase-resolved reconstruction, volumetry and ejection fraction are not studied, and nothing below should be read as evidence about them.

The opportunity is large because the examinations already exist. Chest CT is acquired in very large numbers, and lung cancer screening alone has made it a population-scale examination;[1] the heart is in every one of those scans. Automated methods increasingly read it: fully automated deep-learning coronary calcium scoring now runs on non-gated low-dose chest CT,[2] and its output depends on the acquisition parameters of the scan it is given.[3] In the gated setting, deep learning is also used to remove cardiac motion from the images themselves.[4]

A method of that kind makes a claim about a heart that was never synchronised to the scanner. The obvious way to check such a claim is to compare it with the rate the scanner recorded, and that comparison is not available. Of the 18 series whose images I analysed, none records the rate by any of 11 routes through which a CT acquisition can write one.

Rather than assess any particular method, I ask what the acquisition can contain. Recovering timing from the image itself is not a new idea in CT. Reconstruction correlated with a recorded electrocardiogram came first,[5] and was then replaced by a signal read out of the projection data — the kymogram[6] — and later by gating estimated from the reconstructed images themselves.[7] The same idea carried small-animal imaging, where respiratory and cardiac gating without a physiological monitor became routine.[8,9] Every one of those settings has an acquisition that dwells: a cardiac or micro-CT protocol writes many cycles into the data. That line of work largely stops around 2009.

A separate literature treats the heart in non-gated chest CT, but structurally — thrombus, calcification, aortic root, incidental infarct, epicardial fat. Gating inferred from the data rather than from a monitor has continued elsewhere, in MRI[10] and in PET, where it now outperforms device-based gating,[11] but not in helical CT of the chest. I found no indexed report that recovers cardiac timing from a non-gated helical CT, and none that states the condition under which it could be recovered.

This paper sets out the two sampling requirements recovery has to satisfy, shows that their product is fixed by anatomy and reconstruction interval alone, and measures the constant one of them needs for

one estimator on simulated traces. It then evaluates the requirements against the installed base from acquisition headers, and tests recovery on images under a criterion frozen in advance. That image test is exploratory: no public acquisition I could find pairs a reconstructed helical chest series with a recorded rate, so a self-consistent estimate cannot be confirmed as a correct one, and I report which further measurements separate the two. A simulation in the supplementary material asks what uncertainty a learned estimator declares outside its training range. The contribution is a way of connecting acquisition parameters to the interpretation of an estimate: a framework for choosing the acquisitions on which a future period or phase estimator could be tested, and for reading what its output means. It is not a limit on the cardiac information a CT can hold, and it is not a demonstration of clinical usefulness.

2. Theory

2.1 The z axis of a helical scan is a time axis

In a helical acquisition the table advances at a constant speed S while the gantry rotates, so the axial coordinate of the reconstructed volume and the time of acquisition are related by $z = St$. The scale is known from the header, and the *nominal* acquisition time it gives is the same for every voxel at a given z whatever its row or column: to that approximation, a slice is a slice of time as much as of anatomy. How far the reconstructed value at a voxel departs from that nominal instant is the subject of the assumptions below.

A structure that moves periodically with period T therefore writes that period into the volume as a *spatial* period along z,

$$\lambda = S\,T.$$

For the heart this is the wavy left mediastinal silhouette familiar from a coronal reformat. Below the heart the same silhouette continues as the descending aorta against the left lung, extending the length over which the wave is available.

Two assumptions are carried by $\lambda = ST$ and neither is tested in this study. The first is that the table speed is constant across the analysed span and that the header reports it correctly. The second concerns reconstruction. Each reconstructed slice is formed from projections spanning a finite gantry interval and, on a multi-row detector, from interpolation along z, so the border position recorded at a given z is an average over a temporal footprint rather than an instantaneous position. Such averaging attenuates the periodic component and can displace its apparent phase, by an amount that depends on the reconstruction algorithm, the collimation and the interpolation used. The relation $z = St$ describes the acquisition; it is not a guarantee that the periodic displacement reaches the voxels undistorted, and the simulations of section 3 model the trace rather than the reconstruction that produced it.

2.2 The recoverability window

Whether λ can be recovered is a sampling question with two sides that pull against each other.

The scan must be slow enough that more than one cycle is written across the structure. If the available craniocaudal extent is L, the number of cycles written is $N = L/\lambda = L/(ST)$, and measuring a period requires N to exceed some Nmin. This places an upper limit on S.

The scan must be fast enough that each cycle is spread across enough reconstructed samples to be resolved. With reconstruction interval dz, the samples per cycle are $n = \lambda/dz = ST/dz$, and this must exceed some $n$min. This places a lower limit on S.

The value of $n$min is a judgement and is the one such number in this analysis. Two samples per cycle is the formal floor and is not usable, because a two-sample sinusoid is a straight line to any estimator with noise in it. I take eight throughout, and the sweep of section 3 covers 8, 16 and 32 samples per cycle so that the effect of the choice on Nmin can be read off rather than assumed. Nothing in the budget identity below depends on it.

Together they define a window in table speed, in closed form, for a given L, dz and heart rate. The trade-off itself is old — pitch has always been chosen against longitudinal sampling[12,13] — but the quantity being traded here is cardiac timing rather than spatial resolution. The acquisition side of the window comes entirely from the DICOM header of any helical scan: table speed (0018,9309), or equivalently spiral pitch factor (0018,9311) with total collimation (0018,9307) and revolution time (0018,9305). The other two quantities are not in the header. L is an assumption about which structure carries the motion, and T is the patient's period, which none of the series studied here records. The window is therefore evaluated under stated assumptions about both, which is why section 4 reports it at two values of L.

One ambiguity has to be closed explicitly. The spiral pitch factor of the IEC convention is table travel per rotation divided by the *total* collimated width, whereas the convention in use on four-row scanners divided by a *single* row width; the two differ by the number of rows. The implementation converts between them rather than accepting either silently.

2.3 The budget identity

Multiplying the two quantities removes the heart rate and the table speed at once:

$$N \times n = (L/ST) \times (ST/dz) = L / dz.$$

The product of cycles observed and samples per cycle is fixed by the anatomy and the reconstruction interval alone. It does not depend on pitch, on rotation time, or on how fast the heart is beating. A protocol cannot acquire more cardiac timing by running faster or slower; it can only divide a fixed

budget between seeing more cycles and resolving each one better. Making a scan slower buys cycles at the cost of resolution within a cycle, and making it faster does the reverse.

This is why the question has a definite answer for each protocol, and why the answer is computable before any image exists (figure 1). It also locates the one quantity the condition still needs: the window is bounded by *N*min, and *N*min is not given by the geometry. Section 3 measures it.

3. How many cycles are needed

3.1 Simulation design

How precisely a frequency can be estimated from a finite record is a classical question, and the answer depends on the observation length, the signal-to-noise ratio and the estimator.[14] What that theory does not give is the number of cycles at which a particular estimator, reading a particular signal, starts to succeed in practice. *N*min is a property of the estimation problem — of noise, of how well the cardiac border can be located against lung and mediastinum, and of how far a real rhythm departs from a fixed period — so it was measured rather than assumed. A periodic border trace was generated with a prescribed period, a polynomial baseline standing for anatomy, additive noise, and beat-to-beat variability, and the period was estimated from it. Recovery was scored against the prescribed period at a tolerance of 5 per cent, and *N*min for a condition was taken as the smallest number of written cycles at which at least 90 per cent of 200 trials succeeded.

The sweep covered 144 combinations, the full grid of 4 noise levels from 0.05 to 0.4 of the wave amplitude, 4 baseline strengths from zero to 2.0 times it, 3 sampling densities of 8, 16 and 32 samples per cycle, and 3 degrees of beat-to-beat variability up to 0.1 of the period. The central condition against which the headline is quoted is noise 0.1, baseline 1.0, 16 samples per cycle and variability 0.05. Reporting the whole grid rather than the central cell is what makes the value a plateau rather than a point.

3.2 How far *N*min moves with the estimator, and with the conditions

Two estimators were run over the same traces: one matched to the generating model, and one using only the fundamental of the trace. The matched estimator reaches 1.5 cycles and the fundamental estimator 2.5 cycles. The remainder of this paper uses 2.5, because the matched value assumes knowledge of the waveform that no real trace supplies, and a bound should be stated at the value a usable estimator can attain.

The figure is stable across the sweep (figure 2): 116 of the 144 cells return exactly 2.5, and the worst cell anywhere in the sweep returns 3.5.

4. Where real protocols fall

4.1 Half of the public archive cannot be asked

Evaluating the condition needs only header fields, but they have to be published. Of 13 public collections probed in The Cancer Imaging Archive[15] for acquisition parameters, 6 publish them and 7 do not. For the latter the question cannot be put at all — not because the scans fail the condition, but because nothing in the archive says what the table was doing.

This is worth stating plainly because it bounds any future study of the same kind, including the validation of motion-correction methods that this paper is about. A collection that omits table speed, pitch, collimation and rotation time cannot support a claim about what its scans could or could not have recorded.

4.2 Table speeds and thresholds

The 192 series come from 94 patients in 5 of the 6 collections that publish the parameters, with at most 40 series taken from any one of them so that a single large collection cannot set the distribution. 4 reach that cap and the fifth contributes 32. Table speed runs from 30.0 to 158.8 mm/s, a factor of 5.3, with a median of 55.0 mm/s.

Because faster hearts write more cycles into the same distance, each protocol has a *lowest* rate it can record. Over the cardiac border alone that threshold has a median of 69 bpm, ranging from 38 to 198 bpm, and 128 of 192 series — 67 per cent — have a threshold below 100 bpm. Using the longer extent available along the descending aorta, the median threshold falls to 28 bpm and 100 per cent of series qualify.

Under the assumptions used here, most of the installed base clears the threshold (figure 3). Because the patients' own rates are not recorded, that is a statement about what those protocols could record, not a count of acquisitions that did satisfy the condition. It is enough to make section 5 worth performing.

The exceptions are instructive. A wide-detector protocol at 691.2 mm/s has a threshold of 864 bpm, and a dual-source high-pitch protocol at 1474.6 mm/s a threshold of 1843 bpm. Both are far above any physiological rate: at the assumed structure extent, and at any physiological rate, these acquisitions cross the heart so quickly that less than one full cycle is written, so no complete repetition is present in the record. They fail the cycle-count criterion of this paper by the widest margin in the survey.

5. Meeting the threshold does not produce a period

5.1 A cohort under a frozen criterion

The cohort comprised 18 series, of which 17 yielded a volume the extraction could read; the remaining 1 is reported as a technical failure and enters no count. They are drawn from the same 192 series evaluated in section 4 — all 18 of them — one series per patient, so 18 patients contribute one each and no patient appears twice. 3 of them were the pilots on which the extraction was developed, and they are retained rather than dropped because removing them after the fact would be a choice made with the outcomes visible.

A series was admitted when the joint fit spanned at least $N\min = 2.5$ cycles, was not pinned at either edge of the physiological band, drew on at least three usable coronal levels, moved by no more than 10 bpm when any one level was dropped, and left a median residual below the amplitude it had fitted. Those thresholds were written down before any series beyond the three pilots was fitted.

One detail of that first condition matters and is easy to miss. The implementation counts cycles over the z span actually analysed, not over L, the extent of the structure that section 2 says can carry the motion. For this cohort the two are close: the analysed spans run from 256 to 654 mm with a median of 324 mm against an aortic extent of 300 mm, and the same 16 series clear $N\min$ either way. It is stated here because for the one series in section 5.4 the two differ by a factor of two.

What the fit reads is a one-dimensional trace of that border against z (figure 4). At each slice of a coronal row, the candidates are the runs above -300 HU that cross the row's midline and are at least 25 mm wide, and the border is the left edge of the chosen run. The border is followed rather than chosen independently on each slice: tracking begins in the middle of the scan where the mediastinum is widest, each slice takes the candidate nearest to the previous one, and a slice with no candidate within 6 mm is left empty rather than filled with the nearest available structure. 6 rows are attempted, at fractions 0.45 to 0.7 of the image height, and every one that yields a trace is reported rather than the clearest. A row is kept when the tracker returns a border on at least 64 slices; rows that fail that test are rows where no run above the lung threshold crosses the midline widely enough often enough, the mediastinum not being equally wide at every height in every patient. Across the cohort 3 to 6 rows survive, and the criterion below requires at least three.

The period is fitted to all levels at once. Writing $u = (z - z_0)/\lambda$, with $\lambda = ST$ the spatial period of section 2.1, and $v = (z - z_0)/\text{span}$, each level is modelled as a sinusoid on a quartic baseline, $a \cos 2\pi u + b \sin 2\pi u + c_0 + c_1 v + c_2 v^2 + c_3 v^3 + c_4 v^4$, whose coefficients are solved by least squares at each trial period; amplitude and phase are therefore free per level while the spatial period λ is common, because there is one heart and because in a helical scan the nominal acquisition time is a function of z alone, so every level shares it to the same approximation (section 2.1). The joint cost at a period is the sum over levels

of the residual sum of squares, each weighted by the reciprocal of that level's variance, so a level with a large anatomical swing does not outvote the others. The spatial period is taken as the minimiser over 600 points spanning 40 to 120 bpm at the header's table speed; that band is physiological, and is never set by what the data prefer.

What that criterion measures is self-consistency. Leave-one-out movement and residual size both ask whether the fit holds still; neither compares the fitted rate with the patient's. No series here records one (section 5.4), so correctness was not available to be tested, and the criterion should not be read as if it had been. The count that follows is therefore a count of series meeting a self-consistency criterion. It is neither a count of correct estimates nor a count of incorrect ones, and it is not an accuracy.

3 of the 17 series met it. The header-level condition predicted 13 of them to be recoverable at a ceiling of 100 bpm, and 2 of those 13 were admitted.

The pre-specified summary was the agreement between prediction and outcome, not the admission rate. The two agreed on 5 of 17 series, 29 per cent.

That number needs reading carefully, and I state it descriptively rather than as a test. The prediction is a header-level statement about whether a rate below 100 bpm could in principle be written; the outcome is a self-consistency verdict that never sees a true rate. They are not two measurements of the same thing, so their disagreement is not evidence that either is wrong, and I attach no null distribution to it. With 17 series and two quantities that are not measurements of the same thing, the figure is not strong enough to establish a relation between criterion and acquisition parameters, nor the absence of one; I report it because it was the pre-specified summary. What the admissions are responding to is examined directly in sections 5.3 and 5.4, which is the more informative route.

5.2 How real traces differ from the simulated ones

The border traces of the real series do not merely fit worse than simulated ones, they differ in kind. Comparing 85 real traces with 85 simulated ones, 88 per cent of the real signal sits in the smooth baseline the fit removes, against 12 per cent of the simulated signal, and the spread of band power across levels is 7.5 times wider in the real traces (0.625 against 0.083). A real trace is mostly anatomy, and how much of it is anatomy varies from level to level in a way the simulation never produced.

5 properties were measured on both populations: the share of the signal removed by the quartic baseline, the fraction of the remaining power inside the physiological band, that band's enrichment over the rest of the spectrum, the rate of large slice-to-slice steps, and the drift of the wave's amplitude across the scan. Two of the five separate the populations sharply. The baseline share is the one quoted above. The other is the amplitude drift: a real trace's wave grows or shrinks by a median factor of 1.47 from one end of the scan to the other, against 1.04 in simulation, so the amplitude the fit assumes to

be constant is not. Large steps are absent from the median trace in both populations but reach 2.0 per hundred slices at the ninetieth percentile of the real ones, against 0.0 in simulation.

These are differences, not causes. I did not put any of them back into the simulator to see whether the failure reappears, and this section should not be read as having excluded them. It establishes that the signal the estimator was designed for is not the signal it is given, and section 5.3 reports a property of the objective that is measured rather than inferred.

5.3 The objective is nearly flat, and the criterion never looked

The criterion asks whether the argmin moves. It does not ask whether the cost surface has a minimum worth finding. Those are different questions whenever the surface is flat, and the following measurement was not part of the frozen analysis; it was added after the fact, for the reason given in section 5.4.

Measure the depth of the minimum as the median cost across the whole physiological band divided by the cost at the period the fit returned. A value of 1.00 would mean the chosen period fits no better than an arbitrary one.

Across the cohort the depth is 1.072 at the median of the rejected series, ranging from 1.012 to 1.465. The 3 admitted series lie inside that range, at 1.106 to 1.142: the period chosen in an accepted series reduces the weighted residual by about a tenth relative to a typical period in the band, which is what the rejected series do as well. On this measure the admissions are not distinguishable from the rejections.

They are also not consensus (figure 5). In each of the 3 admitted series the usable coronal levels — 5, 5 and 6 of them — fitted independently, prefer rates whose spread, the largest minus the smallest within that series, is 44 to 59 bpm. The joint fit imposes one period on levels that individually disagree by half the physiological range, and returns their weighted compromise.

What follows from this is about the criterion, and I am careful about how far it goes. Two things are measured: the minima the criterion accepted are no deeper than those it rejected, and the levels it combined did not agree. Neither measurement uses a true rate, so neither can show that the three admitted rates are *wrong*. What they do show is that the criterion admitted them without evidence that they are right, which is a different and weaker claim than the one a reader would take from the word "recovered".

A natural explanation of how that happens is that a flat objective leaves the minimiser unpinned, so removing one level moves it very little, and the leave-one-out test then reports stability that reflects the shape of the objective rather than the strength of a signal. That account is consistent with everything reported here — shallow minima, disagreeing levels, and stable leave-one-out values occurring together — but co-occurrence is not demonstration, and I have not constructed the

experiment that would separate the flatness from the stability. I therefore offer it as an explanation to be tested, and rest the conclusion on the measurements rather than on the mechanism.

One reading it does weaken is that the extraction simply locks onto the wrong peak. With the best period fitting only about a tenth better than a typical one, there is no peak in these objectives sharp enough for that description to be apt.

5.4 A candidate session with a recorded rate, which could not be used

A criterion that cannot be compared with a true rate can be characterised but not scored for accuracy, so I searched for a session that records one. Of 156 collections in The Cancer Imaging Archive, 102 hold CT and 57 studies are described as a cardiac examination; 2 sessions record a rate in the header, and 1 contains both a recorded rate and a free-running helical series of the chest.

The rate is not where a reader would look for it. `HeartRate (0018,1088)` is empty throughout; the scanner writes the rate into the free text of `ScanOptions (0018,0022)` as a minimum, maximum and average in bpm. A survey restricted to the standard cardiac fields concludes that no public CT records the rate, and is wrong; the count in section 4 was repeated over 11 routes for that reason. In that session a gated acquisition records 75.0 bpm, between 72.0 and 78.0 over its own duration, and a helical series covering 630 mm at 208.7 mm/s begins ten seconds later.

The series could not be used. The extraction did not return the anatomical border the method is defined on. Its tracked position lies at 0.9 per cent of the image width, where a mediastinum in this cohort lies between 41 and 48 per cent, and at one coronal level the value is constant to numerical precision across the whole scan. What structure it locked onto instead is not established. Nothing in the pipeline indicated the failure: the fit converged, the residual test passed and the leave-one-out values clustered, and only two levels departing to the edge of the physiological band kept the series out. The run is reported in the supplementary material with the protocol that was fixed before it, and it does not enter any estimate of recovery accuracy.

Having found it in one series, I measured it in all of them. Of the 17 analysed cohort series, 2 sit within 10 per cent of the image edge and 2 more sit closer to the edge than to the middle. All of the affected series were rejected by the frozen criterion, and the 3 admitted series lie at 41 to 43 per cent. The measurement is made per coronal level and not per series, because a series median near the middle could hide a single level out at the edge, which is what happened on the reference case. All 16 usable levels of the three admitted series lie between 35 and 48 per cent of the image width, and figure 4 draws the border on the image for the one of the three whose border sits deepest. What that establishes is the absence of the specific failure it was written to detect, a border out at the surface. It is not a test of anatomical correctness: a position near the middle of the image is where the mediastinum is, and also where several other structures are. Section 5.5 examines all three admitted

series as images and reports what this check misses. The check that measures this, `analysis/check_border_position.py`, did not exist while the cohort was being analysed.

5.5 What the admitted series were actually measuring

To find out what the three admitted series were measuring, every usable coronal level of each — 16 in all — was drawn as a reformat with its tracked border on it and examined. The images are in the supplementary material, and `analysis/audit_admitted_levels.py` produces them and the numbers below. Neither alters the frozen outcome of section 5.1.

No admitted level is on the skin. None of the 16 lies within 10 per cent of the image edge, so the failure of section 5.4 does not recur here. Looking at the reformats shows a different failure that the position check cannot see. Beyond the lung base, and past the apex, the tracker still returns a border although there is no lung there: both sides of it are soft tissue at about 0 HU, and no lung–mediastinum interface exists. Classifying every returned point by the attenuation on each side — aerated lung one side, soft tissue the other — 2610 of 2997 points, or 87 per cent, sit at such an interface. No level is wholly clean, and on the worst 66 per cent are.

The consequence is not evenly spread, and for one series it is serious. The points that do sit at a lung interface span a shorter stretch of z than the levels as a whole, and the criterion of section 5.1 counted cycles over the whole. Re-counted over the valid stretch at the period each fit returned, 1 of the 3 admitted series falls below $N\mathrm{min}$: its valid stretch is 61 per cent of the span analysed, and the cycles written across it are 1.68 rather than the 2.77 the criterion recorded. That series met the cycle-count condition on paper and does not meet it over the part of the scan where the extraction is on the structure the method is defined on.

Two further properties are visible in the reformats and were not measured before: the traces are sparse, with 88 per cent of slices returning no border at all on the worst level, and the tracked line is smooth where the method expects an undulation — the same thing section 5.3 reads off the objective. For all three admissions, then, the fitted period rests on a partial trace of a partly correct border.

6. Discussion

This study establishes four things, and they rest on evidence of different kinds. The two sampling requirements follow from the geometry of a helical acquisition, and their product is fixed by the structure extent and the reconstruction interval, so table speed redistributes a budget it cannot enlarge. The number of cycles one estimator needs was measured rather than assumed. Applied to 192 real series through their headers, and under stated assumptions about extent and rate, the requirements are satisfiable for most of them. And on images, where no rate is recorded, a criterion that asks only whether a fit holds still admits series whose objective is nearly flat, so such a criterion does not

establish that a period was found — the most transferable of the four, and the one this section returns to last.

Two of those results share a structure that is worth stating once.

In section 5 a stability criterion admitted 3 series whose coronal levels, taken separately, preferred rates spread over 44 to 59 bpm, on cost surfaces whose minima are no deeper than those of the series it rejected. In the supplementary material an estimator trained only inside the recoverable regime reported a standard deviation of 0.045 while achieving 0 per cent accuracy outside it. In both, a diagnostic internal to the method reported confidence in a regime the method could not serve, and in neither did the method's own output indicate a problem.

The difference between them is instructive. The estimator's failure is a training failure and has a training remedy: shown the boundary during training, the same architecture widens its reported uncertainty to 0.239 and reports it where an answer is not available, while reaching 85 per cent accuracy at two cycles — the calibration is what changes, not a rule about when to answer. The criterion's failure is a design failure, and what section 5.3 measures against it is diagnostic rather than decisive: a stability statistic on its own does not say whether the objective it was computed on has a minimum worth being stable about. Reporting the depth of the minimum, relative to the median over the searched band, alongside the disagreement between levels would have made the ambiguity in the three admissions visible where the stability statistic concealed it. It would not have decided them. No acceptance threshold on depth is proposed here and none is validated: doing so would need cases whose rates are known, which is what this study lacks. The depth is something to report alongside a fit, not a test the fit can pass.

None of this weakens the geometry. What follows separates the three positive claims by the strength of the evidence behind each, because they should not be read as one.

The budget identity $N\,n = L/dz$ follows from the definitions and holds for any helical acquisition. It is the firmest thing here.

$N\mathrm{min} = 2.5$ cycles is an empirical threshold, measured for one estimator reading one family of simulated traces at one tolerance and one success rate. It is not a universal limit: the matched estimator of section 3.2 reaches 1.5 cycles on the same traces, and the network of the supplementary material (S4.2), trained on that family, is accurate at two. What the bound supports is a practical screen — under the conditions studied here, an acquisition writing appreciably fewer cycles than this should not be expected to yield a period — rather than a statement about every possible estimator.

The header-level evaluation of section 4 is narrower still than it first appears. Of 192 real chest CT series, 67 per cent have a threshold below 100 bpm at an assumed cardiac extent of 120 mm. Because

the actual rates are not recorded, that is a statement about what those protocols could record for a patient beating fast enough, not a count of acquisitions that did satisfy the condition.

What sections 5.3 to 5.5 add is that clearing the threshold does not by itself produce a period: the objective is nearly flat, the stability the criterion measures is not evidence that a period was found, and on inspection one of the three admissions does not clear the cycle-count condition over the stretch where its extraction is on the intended structure. The bound is a screen, not a promise.

Where this would sit in a route to clinical use. A component of the undulation attributable to cardiac motion would, if it could be separated, carry phase as well as period, its maxima and minima falling on opposite parts of the cycle; that is the hypothesis which makes reading timing out of an existing scan of interest, and it is not tested here. What the present study contributes to that route is narrow and comes early in it. The sampling requirements say which acquisition conditions meet the criterion defined in this paper, and are computable from the header before any image is opened, so they can be used to select the acquisitions on which a future estimator is developed or tested. The image evaluation says what an estimate from such a scan does and does not license: a self-consistent fit is not a validated one, and the depth of the objective and the agreement between levels are the quantities to report alongside it. Everything beyond that needs its own validation and receives none here. Identifying which extremum is end-diastole and which end-systole requires a phase reference this study does not have; phase-resolved reformatting requires that the displacement survive reconstruction well enough to sort slices by, which section 2.1 notes is untested; and volumetry or an ejection fraction requires all of that plus a comparison against an accepted measurement. None of those steps is supported or excluded by the results reported here.

Two consequences follow for anyone who would validate such a method. These data did not permit reference validation: one session in 156 collections of The Cancer Imaging Archive holds both a recorded rate and a free-running helical series, and 7 of 13 collections probed do not publish acquisition parameters at all, so for those the requirements cannot even be evaluated. And the failure mode of the supplementary material — a confident report from outside the regime a model was trained in — exists and is invisible from the model's own output, which is an argument for measuring it in a deployed estimator rather than a finding about any particular one.

7. Limitations

The bound was measured against prescribed periods, and applied where no reference period is recorded. $N\mathrm{min} = 2.5$ cycles was obtained by simulation over 144 parameter combinations, in which the period is prescribed and recovery can be scored against it. It was then evaluated against 192 real series through their acquisition parameters alone, and applied to 17 image series in which no reference period is recorded. Exactly one real reconstructed image in the archive can be scored against a known

period, and on that series the extraction did not return the intended border, so it could not be used. No real reconstructed image in this study is scored against a known period.

A fourth candidate cause was found during this revision. The cohort outcome remains consistent with the acquisitions writing too little of the cycle, with the fit being inadequate on real anatomy, or with cardiac motion departing too far from a fixed period, and this study does not separate those three. To them must be added the extraction not being on the intended structure, which sections 5.4 and 5.5 show for the rejected series and for part of every admitted level respectively. Nothing in the pipeline indicated either. Before this revision the study had no check that it was measuring the right thing.

What the evidence does not support is any claim that no method could recover the period. I can say that this extraction finds no peak worth locking onto in the traces where it is on the mediastinum, and that where it is not on the mediastinum the question was never asked. The result is a property of one border tracker and one joint sinusoidal fit on seventeen series. It does not bear on other ways of reading the same undulation — selecting extremal levels directly, correlating whole reformats, or estimating phase without first estimating a period — nor on any subsequent step such as phase-resolved reconstruction or volumetry, none of which is attempted here.

No reference heart rate exists in the cohort itself. None of the 18 series retrieved for image analysis records the rate by any of the 11 routes I checked: the 10 standard DICOM cardiac fields, and the free text of ScanOptions, where a Siemens cardiac acquisition writes the rate while leaving the standard fields empty. The one session that does record it is not in the cohort, and its rate is read from a gated series acquired ten seconds before the series under test rather than from that series itself; the recorded minimum and maximum, 72.0 and 78.0 bpm, are the scanner's own measure of how far the rate moved during an acquisition of that length, and are carried into the tolerance rather than ignored. The archive search that found it keys on study descriptions naming a cardiac examination, so 1 is a lower bound.

What would settle it, and what would not. Scans of a moving phantom driven at known periods, repeated across a range of table speeds and pitches, would score recovery against truth in real reconstructed images and would separate the acquisition and extraction causes from each other. I did not have access to a scanner and a motion phantom, and this work does not include such a study. A digital phantom carried through helical forward projection and reconstruction would reach part of the same ground, testing the extraction on images rather than on traces and allowing acquisition conditions to be swept; it would not settle the third cause, because the beat-to-beat variability of such a phantom is an assumption supplied by the modeller rather than a property of a patient.

Three analyses here are post-hoc, and their status differs from the rest. The depth measurement of section 5.3, the reference case of section 5.4 and the anatomical audit of section 5.5 were all added after the cohort had been frozen, fitted and written up, in response to editorial feedback on a

presubmission enquiry. The cohort outcome reported in section 5.1 is the frozen one; the audit re-counts cycles for interpretation and does not alter which series were admitted. For the reference case the acceptance interval was fixed in the repository before the images were downloaded, and the repository history shows that order; I make no claim to an independently attested timestamp, because a local file and a local commit are both under the author's control.

8. Conclusions

The axial coordinate of a helical CT is a time axis whose scale the header states, so a periodically moving structure writes its period into the volume as a spatial period. Recovering it needs enough cycles written and enough samples within each, and those two requirements multiply to L/dz: a protocol cannot buy more cardiac timing by changing its speed, only spend a fixed budget differently. That identity follows from the definitions. The constant the window depends on was measured rather than assumed, and it is empirical and estimator-dependent — 2.5 cycles for an estimator using only the fundamental of the trace, 1.5 for one matched to the waveform — so it serves as a practical screen under stated assumptions about the structure extent and the rate, not as a limit on every estimator. Evaluated that way, 67 per cent of 192 real chest CT series could record a rate below 100 bpm at an assumed cardiac extent, and 100 per cent over the aortic extent; because the patients' rates are not recorded, that says what those protocols could do rather than what they did.

Meeting the criterion does not produce a period. Under a criterion frozen in advance, 3 of 17 image series were admitted, and the admissions are not evidence of recovery: their cost minima are no deeper than the rejections', at 1.106 to 1.142 against a rejected median of 1.072, and their coronal levels individually prefer rates differing within a series by 44 to 59 bpm. An anatomical audit of all 16 of their coronal levels adds that 87 per cent of the tracked points sit at a lung interface, and that one of the three, re-counted over the stretch where its extraction is valid, writes 1.68 cycles rather than the 2.77 the criterion recorded. Neither can they be shown to be wrong, because no rate is recorded: the one archive session that records one could not be used, since there the extraction did not return the intended anatomical border and nothing in the pipeline said so. The image evaluation is therefore exploratory, and what it establishes is that a self-consistency criterion does not measure whether a period was found.

Two practical consequences follow. The bound is worth computing, because it comes free from the header, but failing it means only that the recovery rate defining Nmin — 90 per cent of trials within 5 per cent — was not reached under the conditions studied, and not that the acquisition holds no signal or that no estimator could succeed. And a fitted period should be reported together with the depth of the minimum it sits in and with a check that the structure fitted was the intended one. This study began with neither, and adding them is what changed how it reads its own results. The auxiliary

simulation of the supplementary material makes the same point about an estimator's declared uncertainty: a model trained only where recovery was possible reported its smallest uncertainty exactly where it was never right, so that uncertainty too is a quantity to be checked rather than taken at face value. Taken together, the sampling requirements and the image evaluation are useful for the next step rather than being a conclusion about it: they say which acquisitions a future period or phase estimator could be tested on, and what its output would have to show before being believed. Whether a period can be recovered accurately from reconstructed non-gated images is not settled here, and the uses that would follow from it — phase identification, phase-resolved reconstruction, ventricular volumetry — are neither demonstrated nor ruled out.

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The research question, the physical argument of section 2, the study design and the decision rules recorded in the repository are the author's and predate the assistance described here. Generative AI (Claude, Anthropic) was used as a tool in preparing this work: to write and test analysis code, to carry out the survey of the public archive reported in section 5.4, to produce the figures from the frozen result files, and to draft manuscript text from results that already existed. No AI system contributed a research idea, chose a decision rule, or determined a result. Every number in this paper is resolved at build time from a result file produced by code in the public repository, every reference is resolved from its DOI rather than written out, and the author has checked the manuscript against those files and takes full responsibility for its content, including the parts drafted with assistance.

Statements and Declarations

Ethical considerations

This study used only publicly available, de-identified imaging distributed by The Cancer Imaging Archive under Creative Commons licences, and generated no new patient data. Ethical approval was therefore not required and none was sought: secondary analysis of de-identified public data does not constitute human subjects research. Approval for the original acquisitions rests with the contributing institutions and is recorded by the archive. Patient identifiers and series instance identifiers appearing in the released results are the archive's own de-identified ones.

Consent to participate

Not applicable. No participant was recruited or contacted for this study, which analysed images already published in de-identified form.

Consent for publication

Not applicable. The article contains no identifiable individual data.

Declaration of conflicting interest

The author declares the following potential conflict of interest with respect to the research, authorship and publication of this article: the author is the representative of LISIT Co., Ltd., a provider of medical imaging analysis services, and this work was carried out within that company. The company received no funding for the study, played no part in its design, analysis or reporting, and markets no product based on the method examined here. The author declares no competing interest beyond that affiliation.

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Data availability

All analysis code, the frozen decision rules and every result file from which the figures and numbers in this paper are computed are available at https://github.com/Institute-of-One/non-ecg-core, at the tagged release v0.1.1, which accompanies this manuscript, and archived at Zenodo, https://doi.org/10.5281/zenodo.23051189; the concept identifier 10.5281/zenodo.23034327 resolves to the most recent version. A clean copy of that release, unpacked into an empty directory and installed from `requirements-core.txt` alone, rebuilds every number in this paper to the same value and draws every figure that does not show an image; `verify_release.py`, in the release, is the script that checks it. Five figures do show one -- figure 4 in the main text and figures S1 and S4 to S6 in the supplementary material, all of them coronal reformats -- and those need the archive series themselves. No imaging data is redistributed here: the code retrieves it from The Cancer Imaging Archive by series identifier, which requires the full `requirements.txt` and a network connection. The collections used are cited in the text.

References

1. The National Lung Screening Trial Research Team. Reduced Lung-Cancer Mortality with Low-Dose Computed Tomographic Screening. N Engl J Med 2011; 365: 395–409.
2. Kim S, Park E, Ahn C, et al. Performance of fully automated deep-learning-based coronary artery calcium scoring in ECG-gated calcium CT and non-gated low-dose chest CT. Eur Radiol 2025; 35: 7084–7095.

3. Chen Y, Wang Y, Wang Q, et al. Impact of scanning parameters on deep learning coronary artery calcium scoring in non-gated chest CT. BMC Med Imaging 2026; 26.

4. Ren P, He Y, Zhu Y, et al. Motion artefact reduction in coronary CT angiography images with a deep learning method. BMC Med Imaging 2022; 22.

5. Kachelriess M and Kalender WA. Electrocardiogram-correlated image reconstruction from subsecond spiral computed tomography scans of the heart. Med Phys 1998; 25: 2417–2431.

6. Kachelrieß M, Sennst D, Maxlmoser W, et al. Kymogram detection and kymogram-correlated image reconstruction from subsecond spiral computed tomography scans of the heart. Med Phys 2002; 29: 1489–1503.

7. Rohkohl C, Prümmer M, Fahrig R, et al. Cardiac C-arm CT: image-based gating. SPIE Proceedings 2008; 6913: 69131G.

8. Badea C, Hedlund LW and Johnson GA. Micro-CT with respiratory and cardiac gating. Med Phys 2004; 31: 3324–3329.

9. Bartling SH, Stiller W, Grasruck M, et al. Retrospective Motion Gating in Small Animal CT of Mice and Rats. Invest Radiol 2007; 42: 704–714.

10. Larson AC, White RD, Laub G, et al. Self-gated cardiac cine MRI. Magn Reson Med 2003; 51: 93–102.

11. Walker MD, Morgan AJ, Bradley KM, et al. Data-Driven Respiratory Gating Outperforms Device-Based Gating for Clinical ¹⁸F-FDG PET/CT. J Nucl Med 2020; 61: 1678–1683.

12. Wang G and Vannier MW. Optimal pitch in spiral computed tomography. Med Phys 1997; 24: 1635–1639.

13. Kalender WA. X-ray computed tomography. Phys Med Biol 2006; 51: R29–R43.

14. Rife D and Boorstyn R. Single tone parameter estimation from discrete-time observations. IEEE Transactions on Information Theory 1974; 20: 591–598.

15. Clark K, Vendt B, Smith K, et al. The Cancer Imaging Archive (TCIA): Maintaining and Operating a Public Information Repository. J Digit Imaging 2013; 26: 1045–1057.

Figure legends

Figure 1. Cycles written across the heart against samples per cardiac cycle, at 60 bpm, on logarithmic axes. Each protocol is a point; the grey lines are the budget $N\,n = L/dz$ for three reconstruction intervals, and a protocol can move only along its own line. The shaded corner is the region in which both conditions hold, bounded below by $N\mathrm{min} = 2.5$ cycles and on the left by $n = 8$ samples per cycle.

Blue points satisfy both conditions, red points fail at least one. The two 2003 four-row protocols are the acquisitions from which this problem originally arose; the modern chest protocols sit an order of magnitude to the right and below the bound, having spent the whole budget on resolution within a cycle.

Figure 2. Left: the fraction of 200 trials in which the prescribed period was recovered to within 5 per cent, against the number of cycles written, for the matched and the fundamental estimator at the central parameter condition. *N*min is where each curve crosses 90 per cent, marked by the dashed verticals at 1.5 and 2.5 cycles. Right: *N*min obtained independently in each of the 144 cells of the sensitivity sweep. The constant is a plateau, not a point estimate. The regime that degrades it is coarse sampling within a cycle combined with high noise, which is the corner in which the budget identity of section 2.3 says the two requirements are already competing for the same fixed product.

Figure 3. The lowest heart rate each of the 192 real chest CT series could record, computed from its header alone at *N*min = 2.5 cycles, over the cardiac border (*L* = 120 mm) and over the descending aorta (*L* = 300 mm). The vertical line is 100 bpm: series to its left could record any physiological rate above their own threshold.

Figure 4. A coronal reformat from one of the admitted series, at its own aspect — one millimetre along z is one millimetre along x — with the tracked border drawn on it. The border follows the mediastinum between the two lungs, which is the structure the method is defined on. Its position, 43 per cent of the image width here, is what section 5.4 measures for every series.

Figure 5. Left: depth of the minimum, defined as the median cost across the physiological band divided by the cost at the period returned, for the series the frozen criterion rejected and for those it admitted. A value of 1.0, the dashed line, would mean the chosen period fits no better than an arbitrary one. The admitted series lie inside the range of the rejected ones. Right: for each admitted series, the rate each usable coronal level prefers when fitted alone (grey), against the rate the joint fit returns (cross). The joint fit reports a compromise among levels that do not agree.